Selected Projects

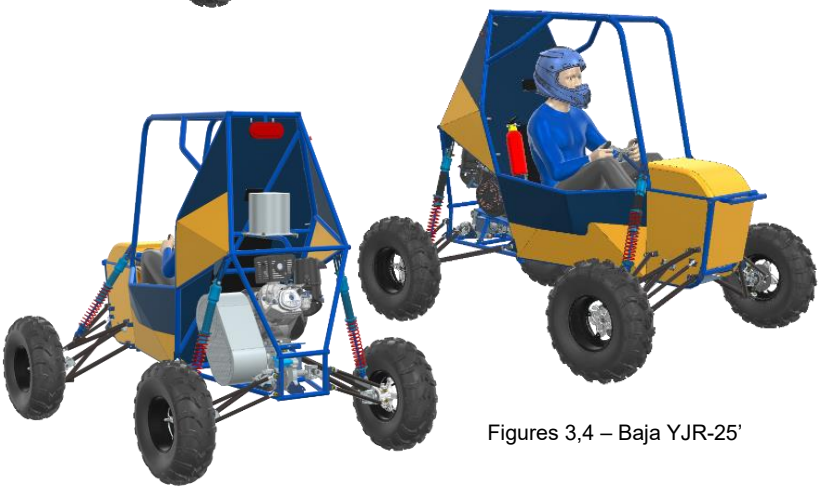
BAJA SAE

- Develop and maintain the full vehicle CAD assembly (Figures 1-4).
- Manage fasteners, ensuring proper hardware is documented and in stock.
- Estimated vehicle weight and calculated center of gravity (CG) virtually in CAD, achieving 2.4% error in CG compared to measured results.
- Manage component weight measurements and maintain material property references.
- System-level motion simulations and analysis (Figure 5, ongoing).
- Verified assembly integration using CAD interference checks and motion analysis.
- Performed FEA of select components using SOL101 linear static analysis in NX Nastran.
- Modeled CAD virtual body panels and graphics.
- Created standardized CAD files for use across multiple subassemblies, improving consistency.
- Provide subsystem leads with NX CAD modeling and simulation guidance.
- 3D printer maintenance and repair (MakerBot, Flashforge).
- Lead the Pumpkin launch subgroup in upgrading, maintaining, and competing in the annual ASME Pumpkin Launch competition.

(CAD: NX 2406/2506/2606)



Figures 1,2 – Baja YJR-26'



Figures 3,4 – Baja YJR-25'

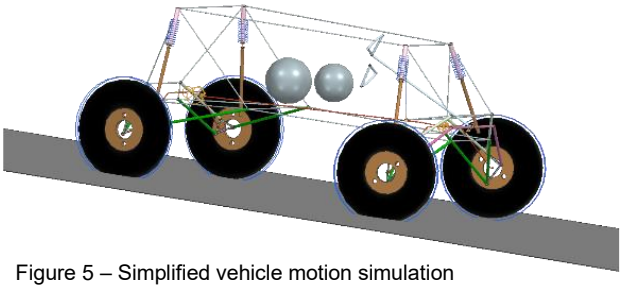


Figure 5 – Simplified vehicle motion simulation

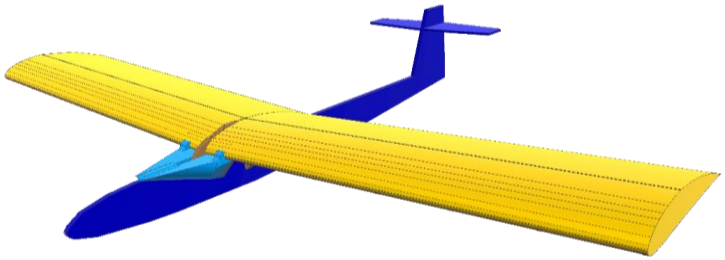


Figure 6 – Low Speed Glider

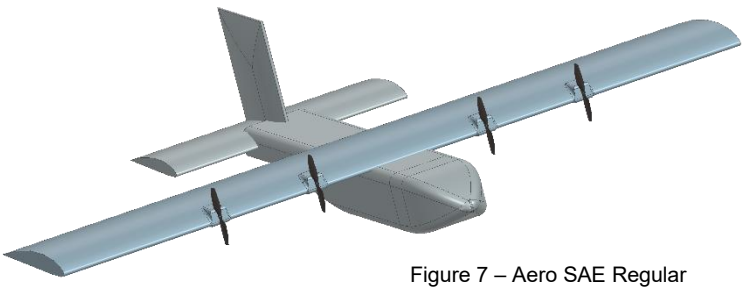


Figure 7 – Aero SAE Regular Class Aircraft

AERO SAE

Low Speed Glider for Mini University Competition

- Laser cut cardboard frame and 3D printed counterweight for variable center of gravity control.
- Achieved first in accurate target flight.
- Used NX 2506 and RDWorksV8 software.

SAE Aero Design Regular Class Aircraft

- Designed and analyzed the airframe for an SAE Aero Design competition aircraft.
- Performed initial aerodynamic calculations and optimization using MATLAB.
- Modeled the complete outer shell in Siemens Design Center / NX, including the wing, fuselage, nose cone, and tail.

Laboratory for Laser Energetics (LLE)

High Precision X-Ray Crystal Reflectivity Tower

- Developed and iterated a precision positioning system using linear, rotational, and solenoid-driven stages for automated X-ray crystal measurements.
- Iteration 1: Single-point crystal scanning system for reflectivity measurements.
- Iteration 2: Expanded the system to automatically scan all points across the crystal.
- Controls programmed in Python.
- Applied GD&T for fabrication by LLE machinists.
- Verified assemblies in Siemens NX and through physical 3D-printed prototypes.
- Created a Maxwell-style kinematic coupling for repeatable positioning of the solenoid-driven stage.
- Designed around readily available laboratory equipment and McMaster-Carr components to minimize custom hardware and simplify integration.
- Developed crystal realignment procedure.

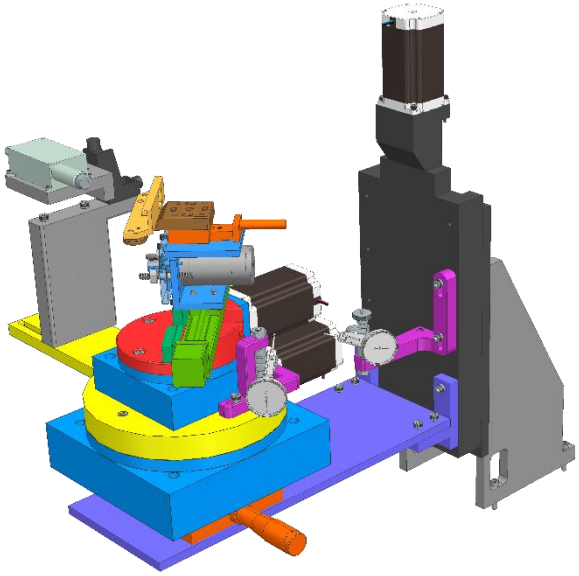


Figure 8 – High Precision X-Ray Reflectivity Tower

Automated X-Ray Source Optical and Filter Shutter

- Developed a modular, 3D-printable frame with interchangeable components for manual or remote X-ray beam blocking and filter activation.
- Programmed a custom Python GUI for remote control.
- Developed a dedicated solenoid control box integrating a power supply, relays, and data-acquisition modules.

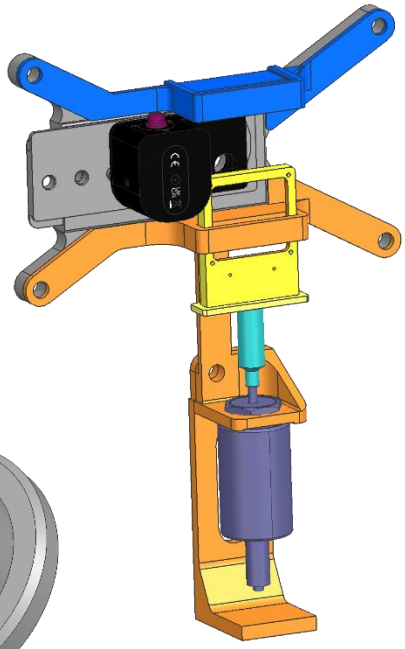


Figure 9 – Optical and Filter Shutter

SLOS-TRXI Photocathode Tool

- Reverse-engineered a severely damaged tool required for photocathode removal from SLOS-TRXI, a high-speed imaging diagnostic instrument.
- Designed and fabricated a functional replacement tool within 24 hours to restore operation.

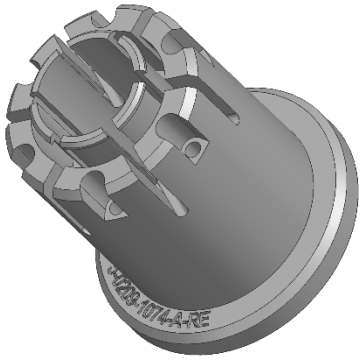


Figure 10 – SLOS-TRXI Tool

Gas-Jet Filter Adapter

- Designed an adapter to position a small filter directly adjacent to a pressurized supersonic gas-jet nozzle for experimental measurements.
- Modeled, 3D-printed, and deployed on experimental gas-jet within 5 hours.

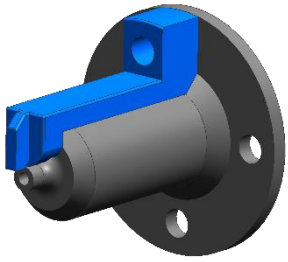


Figure 11 – Gas-Jet Adapter

Other Projects:

Aluminum X-Ray Filters – Laser ablation of aluminum film with UV and IR fiber lasers for X-ray filters.

Optical Laser Alignment for X-Ray Beam – Used various optics to create an alignment system for the X-ray beam.

X-Ray Attenuation Step Wedge Filters – Improved filter application procedure, reducing defect rate by 50% for 0.5 μm size PET film, in making filters for soft X-ray energy distribution characterization inside the OMEGA-60 chamber.

X-Ray Source Viewing System – Developed a digital optical system to visualize an X-ray source in a vacuum chamber.

2026 Coursework Projects

ME241 Mechanics Lab

Deflection of a Bimetallic Strip Lab

Used Siemens NX to simulate thermal deflection of a 6061 aluminum / 1095 steel bimetallic strip and compared the results against experimentally measured deflection. Evaluated discrepancies between the model and physical results, including adhesive effects and boundary-condition assumptions.

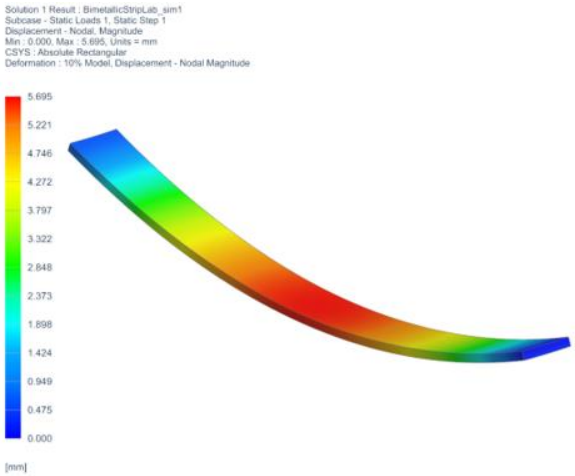


Figure 12 – Bimetallic strip deflection, nodal displacement result from NX simulation with color bar in millimeters.

Drag on an Airfoil Lab

Performed CFD analysis in Siemens NX and compared predicted airfoil drag with experimental data collected in a wind tunnel, calculating the percent difference between results.

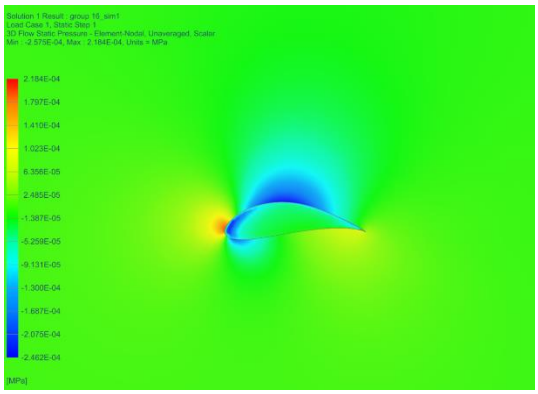


Figure 13 – Pressure contours from the CFD simulation, showing pressure distribution.

Baja SAE Shock Characterization – Course Capstone

Investigated the effect of nitrogen charge pressure on Baja SAE shock performance using an MTS test system and custom-machined fixtures. Evaluated 10-30 psi pressures and found no significant effect on breakaway force. Applied GD&T, CAD, machining, and design of experiments (DOE).

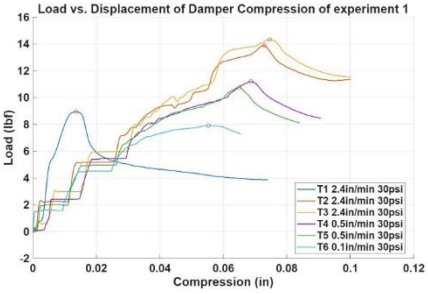


Figure 14 – Breakaway force data from the first experiment displaying steps and inconsistent values for each shock.



Figure 15 – Image of the damper installed on an MTS RT/50 machine.

ME223 Heat Transfer

Luminescence Lifetime Thermometry

Collected and analyzed temperature-dependent luminescence data using oscilloscopes and photomultiplier instrumentation, performing MATLAB-based curve fitting, regression analysis, and sensor sensitivity comparisons with RTDs, thermistors, and thermocouples.

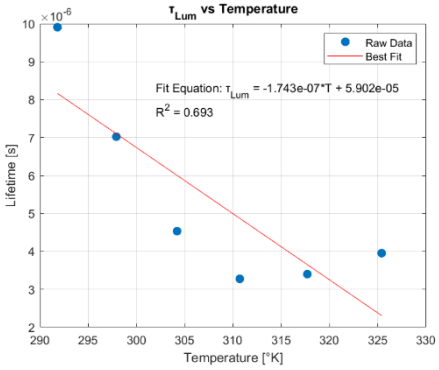


Figure 16 – Luminescence lifetime decreases with increasing temperature.